

Aaliyah Pirani

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EDUCATION

University of Toronto, St. George Campus

Expected: Apr. 2028

Bachelor of Science (Double Major: Computer Science and Quantitative Biology)

GPA: 4.0/4.0

- Relevant Coursework: Software Design, Theory of Computation, Probability/Statistics, Computer Organization
- Awards: TD Scholarship for Community Leadership (\$70,000, May 2024); Howard Ferguson Admission Scholarship (\$12,000, Sep. 2024); DCS Award (\$12,000, Mar. 2026); UofT Scholar (\$1,500, Jul. 2025)

TECHNICAL SKILLS

Languages & Libraries: Python, Java, C, R, pandas, NumPy, PyTorch, NVIDIA Warp, scikit-learn

Tools: Arduino, SolidWorks, Flutter, Microsoft Power BI, Power Automate, PowerApps, FlexiCubes

EXPERIENCE

Undergraduate Research Assistant

May 2026 - Present

MEDCVR Lab, University of Toronto Mississauga

Toronto, ON

- Extended a **GPU-accelerated** physics simulation framework that utilizes **NVIDIA Warp and FlexiCubes** to enable a real-time simulation of elastic material deformations using FEM-based physics approximations
- Compared alternative simulation approaches and evaluated trade-offs in accuracy, stability, and performance
- Implemented **mesh cutting algorithm** to improve real-time interactivity and support future robotic surgical simulation applications such as **laser ablation modeling** on soft-tissue analogs

Computer Science, Math and Statistics Tutor

Feb. 2021 - Present

Private

- Delivered **500+ hours of instruction** to 20+ IB and first-year university students in programming fundamentals, calculus, and probability/statistics, resulting in assessment scores of 85%+ through adaptive instruction

Vice President Internal

Sep. 2025 - Present

University of Toronto Robotics Association

Toronto, ON

- Acted as **primary liaison** between admin team and 6 robotics subteams, facilitating biweekly progress meetings, troubleshooting **technical and organizational** issues and ensuring alignment across independent teams

Data Risk Management Intern

May 2025 – Aug. 2025

Toronto-Dominion Bank

Toronto, ON

- Designed a **low-code Microsoft PowerApp** that automated cross-departmental data ingestion and exchange eliminating manual processing and reducing administrative workload
- Developed and deployed interactive **Power BI dashboards** to aggregate and streamline datasets, improving internal data accessibility and visibility of key reporting aspects

PROJECTS

BearWithMe - Helping Young Minds Master Pronunciation | *Python, Node.js, Raspberry Pi* Nov 2025

- Developed an **embedded learning system** on a Raspberry Pi with microphone and speaker, leveraging Azure Pronunciation Assessment API and ElevenLabs TTS for real-time pronunciation feedback via a physical device
- Built **full-stack features** including a parent-facing progress dashboard and adaptive practice logic, leveraging the Gemini API to dynamically personalize word selection, generate insights, and improve learning consistency

SnackOverflow - Find Your Next Meal | *Java, RestAPI, Git* Nov. 2025 - Dec. 2025

- Integrated core application workflows with Java and Swing, contributing to a **multi-feature desktop app** that supports recipe search, viewing, and saving through clean architecture while adhering to SOLID principles
- Connected **REST APIs** and a **MongoDB database** using Java, enabling real-time recipe retrieval and persistent storage for saved items, improving usability and session continuity

My Very Own Neural Network | *Python, NumPy* Jun. 2025 - Aug. 2025

- Implemented a **feedforward neural network** from first principles in Python and NumPy, manually coding forward propagation, backpropagation, gradient descent, and sigmoid activation without ML frameworks
- Trained and evaluated the model on the MNIST dataset, achieving **93% classification accuracy**, while experimenting with weight initialization, loss optimization, and model evaluation metrics